EAB 770 - Chapter 3

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1 Chapter 3: Sets of 2×2 tables

1.1 Outline

- Mantel-Haenszel test
- Measures of Association
  - Common Odds ratio
  - Breslow-Day test
  - Zelen's Test

2 Mantel-Haenszel Statistic

2.1 Sets of 2×2 tables

- Consider a multi-location RCT
  - Different locations will have different contingency tables
  - We want to consolidate **all** data and account for the location effect in testing for the association between the treatment and outcome.

Table 3.1 Respiratory Improvement

Center	Treatment	Yes	No	Total
1	Test	29	16	45
1	Placebo	14	31	45
Total		43	47	90
2	Test	37	8	45
2	Placebo	24	21	45
Total		61	29	90

2.2 Stratified Analysis

- Stratified analysis examines the association between two variables while adjusting for the effect of other covariates.
  - The variable that define the strata may be an explanatory variable, location, or classification variable.
  - Each table corresponds to one level of the explanatory variables (stratum)

2.3 Mantel-Haenszel Test

Recall the Mantel-Haenszel Chi-Square statistic

$$Q = (n_{11} - m_{11})^2 / v_{11}$$

where m_{11} is the expected value of the n_{11} and v_{11} is the variance of n_{11} based on the hypergeometric distribution assumption.

- If there are q 2×2 tables, each table will have a value for Q.

2.4 Mantel-Haenszel Test

The Mantel-Haenszel test assess the overall association of group and response after adjusting for the strata. If n_{j11} is the n_{11} value of the j^{th} contingency table, the Mantel-Haenszel statistic, Q_{MH} can be written as:

$$Q_{MH} = \frac{\left[\sum_{j=1}^{q} n_{j11} - \sum_{j=1}^{q} m_{j11}\right]^2}{\sum_{j=1}^{q} v_{j11}}$$

> **Note**
>
> Q_{MH} is often called an **average partial association statistic**.

2.5 Mantel-Haenszel Statistic

- The statistic Q_{MH} relies on detecting patterns of the differences of proportions in each table
- The statistic Q_{MH} approximately follows a chi-square distribution with one degree of freedom if the combined random sample sizes are large.
  - Individual cell counts and isolated table sample sizes may be small.
  - The *Mantel-Fleiss criterion* can be used to check if the chi-square assumption holds. The MF score should be greater than 5.

2.6 Mantel-Haenszel Test: SAS

- The `cmh` option in the `tables` statement produces the Mantel-Haenszel statistic values. There will be three Mantel-Haenszel values included in the output.
- Nonzero Correlation
- Row Mean Scores Differ
- General Association

> **Note**
>
> For 2×2 tables, the scores and degrees of freedom do not differ for these three MH-statistics. These measures will be distinguished later in the course.

2.7 Mantel-Haenszel Test: SAS

- The `mf` option in the `tables` statement produces the Mantel-Fleiss criterion. Recall that the Mantel-Fleiss criterion (MF) should be greater than 5 for the chi-square distributional assumption to be valid.

2.8 Example

The table shows the results of a study about how general daily stress affects one's opinion on a proposed new health policy. Participants were sampled from urban and rural areas.

- Is there enough evidence that there is an association between the opinion and stress levels of residents after controlling for residential area? Use =0.05 as the significance level.

Table 3.3 Health Policy Opinion Data

Residence	Stress	Favorable	Unfavorable	Total
Urban	Low	48	12	60
Urban	High	96	94	190
	Total	144	106	250
Rural	Low	55	135	190
Rural	High	7	53	60
	Total	62	188	250

2.9 Example

The table shows the results of a study about how general daily stress affects one's opinion on a proposed new health policy. Participants were sampled from urban and rural areas.

- Is there enough evidence that there is an association between the opinion and stress levels of residents after controlling for residential area? Use =0.05 as the significance level.

2.9.1 SAS Code

```
data stress;
input region $ stress $ outcome $ count @@;
datalines;
urban low f 48 urban low u 12
urban high f 96 urban high u 94
rural low f 55 rural low u 135
rural high f 7 rural high u 53
;
proc freq order=data data=stress;
```

```
weight count;
tables region*stress*outcome / all chisq cmh (mf) nocol nopct;
run;
```

1. `CMH` includes the Mantel-Haenszel statistics. `MF` gives the Mantel-Fleiss condition;

2.9.2 R Code

```r
library(vcdExtra)
```

```
Loading required package: vcd

Loading required package: grid

Loading required package: gnm


Attaching package: 'vcdExtra'

The following object is masked from 'package:dplyr':

    summarise
```

```r
dat  <- expand.grid(Outcome = c("Favorable","Unfavorable"),
                    Residence=c("Urban","Rural"),
                    Stress = c("Low","High")
                  ) |>
mutate(count = c(48,12,55,135,96,94,7,53))

xtabs(count ~  Stress + Outcome + Residence  ,data=dat) -> dat_cc

CMHtest(dat_cc,
        strata="Residence",
        types="ALL",
        overall=T)
```

```
$`Residence:Urban`
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
    in stratum Residence:Urban

                 AltHypothesis  Chisq Df       Prob
general    General association 16.155  1 5.8367e-05
rmeans  Row mean scores differ 16.155  1 5.8367e-05
cmeans  Col mean scores differ 16.155  1 5.8367e-05
cor        Nonzero correlation 16.155  1 5.8367e-05


$`Residence:Rural`
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
    in stratum Residence:Rural

                 AltHypothesis  Chisq Df      Prob
general    General association 7.2724  1 0.0070022
rmeans  Row mean scores differ 7.2724  1 0.0070022
cmeans  Col mean scores differ 7.2724  1 0.0070022
cor        Nonzero correlation 7.2724  1 0.0070022


$ALL
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
    Overall tests, controlling for all strata

                 AltHypothesis Chisq Df       Prob
general    General association 23.05  1 1.5783e-06
rmeans  Row mean scores differ 23.05  1 1.5783e-06
cmeans  Col mean scores differ 23.05  1 1.5783e-06
cor        Nonzero correlation 23.05  1 1.5783e-06
```

2.9.3 Answer

$\chi^2 = 23.05$, $df = 1$, $p < 0.0001$. There is strong evidence that the stress level is associated with the opinion outcome.

2.10 Example

The table shows the results of an RCT for a test treatment designed for respiratory infections. Is there a significant association between the treatment and the outcome after controlling for the center?

- Is the Mantel-Fleiss criterion satisfied?

```
, , Center = 1

         Improvement
Treatment Yes No
  Test     29 16
  Placebo  14 31

, , Center = 2

         Improvement
Treatment Yes No
  Test     37  8
  Placebo  24 21
```

2.11 Example

The table shows the results of an RCT for a test treatment designed for respiratory infections. Is there a significant association between the treatment and the outcome after controlling for the center?

- Is the Mantel-Fleiss criterion satisfied?

2.11.1 SAS Code

```
data resp;
length treatment $7.;
input treatment $ improvement $ center $ count;
datalines;
Test Yes 1 29
Placebo Yes 1 14
Test No 1 16
Placebo No 1 31
Test Yes 2 37
Placebo Yes 2 24
Test No 2 8
Placebo No 2 21
```

```
;


proc freq data=resp order=data;
weight count;
tables center*treatment*improvement / cmh (MF) nocol norow nopct;
run;
```

2.11.2 R Code

```r
dat  <- expand.grid(Improvement = c("Yes","No"),
                    Center=c(1,2),
                    Treatment = c("Test","Placebo")
                  ) |>
mutate(count = c(29,16,37,8,14,31,24,21))

xtabs(count ~  Treatment+Improvement + Center, data=dat) -> dat_cc
dat_cc
```

```
, , Center = 1

         Improvement
Treatment Yes No
  Test     29 16
  Placebo  14 31

, , Center = 2

         Improvement
Treatment Yes No
  Test     37  8
  Placebo  24 21
```

```r
CMHtest(dat_cc,
        strata="Center",
        types="ALL",
        overall=T)
```

```
$`Center:1`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
```

```
    in stratum Center:1

                 AltHypothesis  Chisq Df      Prob
general    General association 9.9085  1 0.0016452
rmeans  Row mean scores differ 9.9085  1 0.0016452
cmeans  Col mean scores differ 9.9085  1 0.0016452
cor        Nonzero correlation 9.9085  1 0.0016452


$`Center:2`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    in stratum Center:2

                 AltHypothesis  Chisq Df      Prob
general    General association 8.5025  1 0.0035465
rmeans  Row mean scores differ 8.5025  1 0.0035465
cmeans  Col mean scores differ 8.5025  1 0.0035465
cor        Nonzero correlation 8.5025  1 0.0035465


$ALL
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    Overall tests, controlling for all strata

                 AltHypothesis  Chisq Df       Prob
general    General association 18.411  1 1.7807e-05
rmeans  Row mean scores differ 18.411  1 1.7807e-05
cmeans  Col mean scores differ 18.411  1 1.7807e-05
cor        Nonzero correlation 18.411  1 1.7807e-05
```

2.11.3 Answer

- The Mantel-Fleiss criterion was determined to be 36, which means the chi-square assumption holds.
- $Q_{MH} = 18.4, p < 0.0001$, there is strong evidence of an association between the treatment and improvement.

3 Measures of Association

3.1 Common Odds Ratio

- An odds ratio can be calculated for each stratum based on the 2x2 table values.
  - What if there are multiple strata?
- If the odds ratios are expected to be the same over all strata (homogenous), then we can calculate a common odds ratio ψ.

3.2 Common Odds Ratio

- The Mantel-Haenszel estimator $\hat{\psi}_{MH}$ can approximate the common odds ratio such that

$$\hat{\psi}_{MH} = \frac{\sum_{j=1}^{q} n_{j11} n_{j22}/n_j}{\sum_{j=1}^{q} n_{j12} n_{j21}/n_j}$$

> **Note**
>
> We can calculate confidence intervals (asymptotic and exact) for common odds ratios!

3.3 Common Odds Ratio: Confidence Intervals

There are two confidence intervals for the common odds ratios provided by SAS: The Mantel-Haenszel and Logit confidence limits.

3.3.1 Mantel-Haenszel

- based on an estimate of the variance of $\log \psi_{MH}$
- Not sensitive to small sample sizes

3.3.2 Logit

- based on weighted regression estimates
- reasonable but requires adequate sample sizes, i.e. $n_{hij} \geq 5$

3.4 Homogeneity of Odds Ratios

> **Warning**
>
> It is important to remember that the common odds ratio assumes the odds ratio is uniform across all strata!

The Breslow-Day test or the Zelen's test can be used to test if the homogeneity assumption is satisfied.

3.4.1 Breslow-Day test

- Null hypothesis: All strata are homogeneous.
- Assume the null value of the odds ratio is ψ_{MH} for each stratum
- Calculate the expected values of the cells based on ψ_{MH}

$$Q_{BD} = \sum_{h=1}^{q} \sum_{i=1}^{q} \sum_{j=1}^{q} \frac{(n_{hij} - m_{hij})^2}{m_{hij}}$$

which approximately follows a chi-square distribution with $(n-1)$ degrees of freedom.

3.4.2 Zelen's Test

Zelen's test is similar to the Breslow-Day test, but uses exact methods.

- Recall: Fisher's Exact Test
- calculates the probability of all possible sets of tables.
- p-value is the sum of all table probabilities that are lower than the probability of the observed table.

3.5 Homogeneity of Odds ratios

If the Breslow-Day test or Zelen's test is significant, then the common odds ratio might not be an appropriate measure of association.

- The strata must be analyzed separately (similar to interactions in linear models.)

3.6 Common Odds Ratios: SAS

- The common odds ratios are calculated with the `cmh` option
  - Only includes asymptotic (MH and logit) CI limits.
  - Including the `measures` options yields plots of common odds ratios and stratum-level odds ratios.
- The Breslow-Day test is also included in the `cmh` option
- The `comor` and `eqor` options in the `exact` statement yield the exact common odds ratio and the Zelen's test results, respectively.

Sample SAS code:

```
proc freq;
tables a*b*c/cmh measures;
exact comor eqor;
run;
```

3.7 Example

The table shows the data from a study on coronary artery disease. Investigators were interested in whether electrocardiogram (ECG) measurement was associated with disease status. Gender was thought to be associated with disease status, so investigators post-stratified the data into male and female groups. In addition, there was interest in examining the odds ratios.

Table 3.7 Coronary Artery Disease Data

Sex	ECG	No Disease	Disease	Total
Female	< 0.1 ST segment depression	11	4	15
Female	≥ 0.1 ST segment depression	10	8	18
Male	< 0.1 ST segment depression	9	9	18
Male	≥ 0.1 ST segment depression	6	21	27

3.8 Example

The table shows the data from a study on coronary artery disease. Investigators were interested in whether electrocardiogram (ECG) measurement was associated with disease status. Gender was thought to be associated with disease status, so investigators post-stratified the data into male and female groups. In addition, there was interest in examining the odds ratios.

3.8.1 SAS Code

```
data ca;
input sex $ ECG $ disease $ count;
datalines;
female <0.1 yes 4
female <0.1 no 11
female >=0.1 yes 8
female >=0.1 no 10
male <0.1 yes 9
male <0.1 no 9
male >=0.1 yes 21
male >=0.1 no 6
;
proc freq data=ca ;
weight count;
tables sex*disease / nocol nopct chisq measures;        ①
tables sex*ECG*disease / nocol nopct cmh measures;      ②
run;
```

① Testing for an association between sex and disease status
② Using sex as a stratification variable/strata. `measures` includes common odds ratio plots.

3.8.2 Answer

- Sex was determined to be associated with disease status ($\chi^2 = 6.9, p = 0.0084$)
- Controlling for sex, there is strong evidence that the ECG measurements are associated with disease status ($\chi^2 = 4.5, p = 0.03$).
- The Breslow-Day test yielded a non-significant p-value (p=0.64), hence we can use the common odds ratios
- The MH odds ratio was estimated to be 2.86 (95% CI: 1.1,7.5). We can interpret this value as: The odds of having the disease is 2.86 times higher for patients with a high ECG segment depression compared to those with a low ECG segment depression.

3.9 Exercise

Table 3.8 shows the data from research done by a small company in initiating exercise programs (office/home) in their two locations in a nearby suburbs. After a year, participants and non-participants underwent a cardiovascular stress test to assess their fitness, and their result was recorded as good or not good depending on age-adjusted criteria. The exercise counselor was interested in whether type of program was associated with good test results.

- Calculate the common odds ratio ψ_{MH} and the **exact** 95% confidence intervals.
- Use Zelen's test to test if the odds ratio is uniform in both locations.

Table 3.8 Cardiovascular Test Outcomes

Location	Program	Good	Not Good	Total
Downtown	Office	12	5	17
Downtown	Home	3	5	8
	Total	15	10	25
Satellite	Office	6	1	7
Satellite	Home	1	3	4
	Total	7	4	11

3.10 Exercise

Table 3.8 shows the data from research done by a small company in initiating exercise programs (office/home) in their two locations in a nearby suburbs. After a year, participants and non-participants underwent a cardiovascular stress test to assess their fitness, and their result was recorded as good or not good depending on age-adjusted criteria. The exercise counselor was interested in whether type of program was associated with good test results.

- Calculate the common odds ratio ψ_{MH} and the **exact** 95% confidence intervals.
- Use Zelen's test to test if the odds ratio is uniform in both locations

3.10.1 SAS Code

```
data exercise;
input location $ program $ outcome $ count @@;
datalines;
downtown office good 12 downtown office not 5
downtown home good 3 downtown home not 5
satellite office good 6 satellite office not 1
satellite home good 1 satellite home not 3
;

proc freq order=data;
weight count;
tables location*program*outcome /cmh(mf);   ①
```

```
exact comor eqor;                                                          ②
run;
```

① `MF` should be enclosed in parentheses
② `eqor`: Zelen's test, `comor`: exact confidence interval for the common odds ratio.

3.10.2 Answer

- Mantel-Fleiss criterion is not satisfied (MF=4.65), so be careful with using the asymptotic chi-square test for Q_{MH}.
  - This motivates using exact methods in calculating effect sizes.
- Common odds ratio was estimated to be 5.84 (**EXACT** 95% CI: 1.05, 33.3). There is sufficient evidence at the 0.05 significance level that the program type is associated with the quality of cardiovascular test results.
  - The Zelen's test yielded $p = 1$, which means there is no evidence against homogeneity.